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Quiz 1

Date: August 25, 2025

Time: 1005 - 1025 hrs

Marks: 10

General instructions:

Write the answers in the space provided below each question.

While providing a type for a function, Integer is preferable to Int in most cases.

1. What is the type of the function f defined below?

$f [] = \text{True}$

$f (x:xs) = \text{let } y = f \text{ xs in } x \neq y$

Give three examples of evaluating f on lists of length at least 4.

Double: ?
Its single
Type: $f :: [t] \rightarrow \text{Bool}$

However, it only gives $\neq$ operation on Booleans
(as last $f []$ call gives Bool)

Type $\rightarrow f :: [\text{Bool}] \rightarrow \text{Bool}$

Examples

1) $f [T, T, T, T] = T \neq f [T, T, T]$

$T \neq (T \neq (T \neq (T \neq f [])))$
 $T \neq (T \neq (T \neq T))$
 $T \neq (T \neq F)$
 $T \neq T$
 F

1) $f [T, T, T, T] = \text{True}$

2) $f [T, T, T, F] = \text{False}$

$T \neq (T \neq (T \neq (F \neq T)))$
 $T \neq (T \neq (T \neq F))$
 $T \neq (T \neq T)$
 $T \neq F$
 F

3) $f [T, F, T, F] = \text{True}$

$T \neq (F \neq (T \neq (F \neq T)))$
 $T \neq (F \neq (T \neq F))$
 $T \neq (F \neq T)$
 $T \neq F$
 F

Rough

$T \neq T$
 $T \neq \text{False}$
 True

4

2. Consider the function $f :: [\text{Integer}] \rightarrow [\text{Integer}]$ given below:

$f [] = [20]$

$f (x:xs) = \text{if } x \geq 0 \text{ then } \min x y: ys \text{ else } ys \text{ where}$

$(y:ys) = f xs$

Trace the evaluation of $f [-10, 2, 45, 3, 22, 19, 6, 12, 10, 25]$ by showing the value of each recursive call.

Note that there are 11 calls to f including the original call and $f []$.

